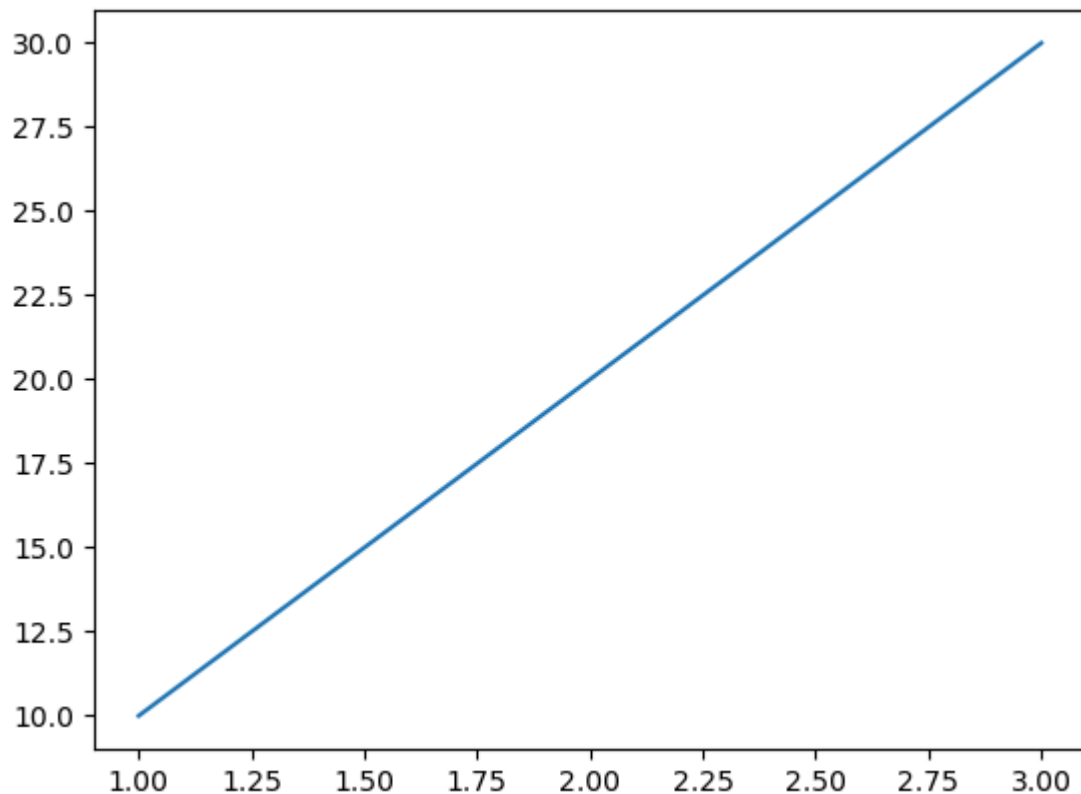
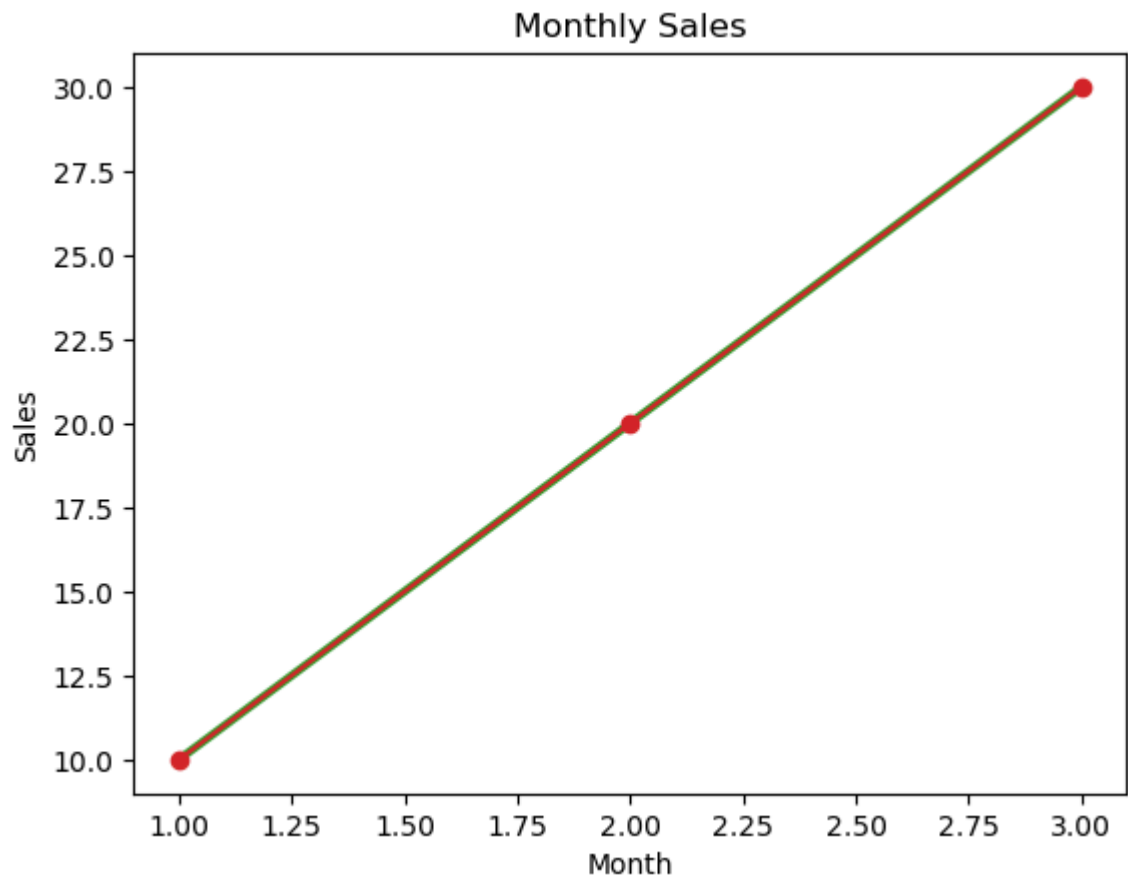


```
In [1]: import matplotlib.pyplot as plt
x = [1,2,3]
y = [10,20,30]
plt.plot(x,y)
plt.show()
```



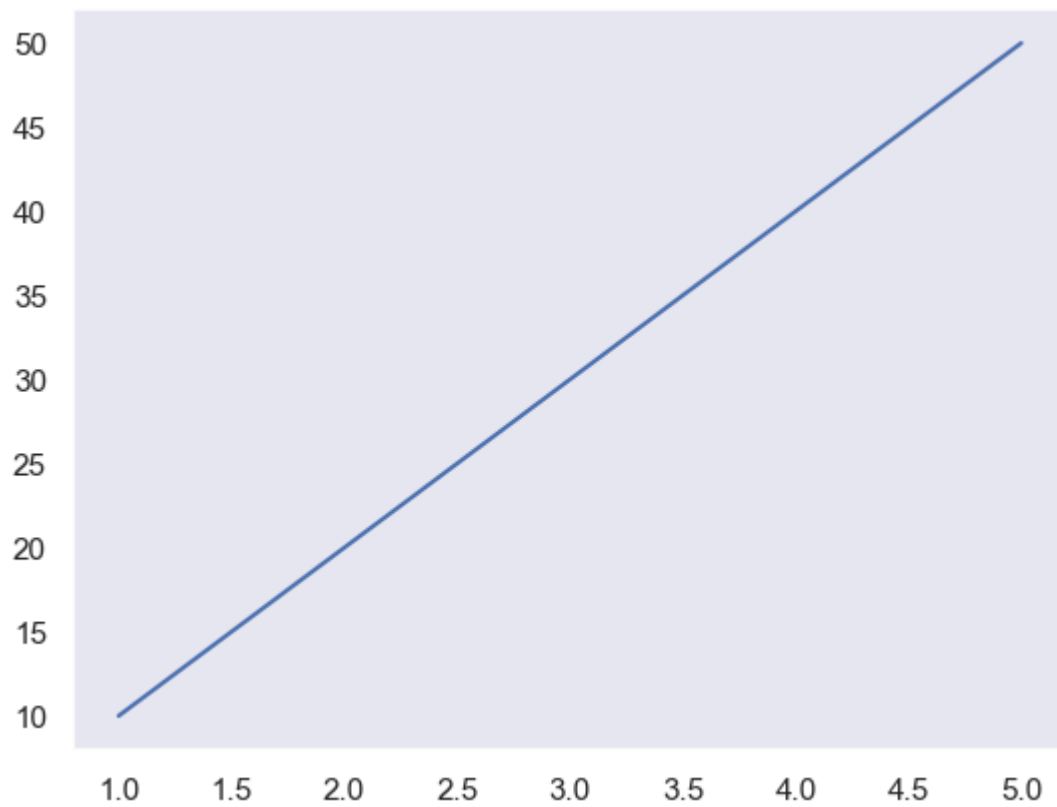
```
In [13]: import matplotlib.pyplot as plt
fig, ax = plt.subplots()
ax.plot([1,2,3],[10,20,30])
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.plot(x,y,color='red')
plt.plot(x,y,linestyle='-.')
#changing linewidth
plt.plot(x,y,linewidth=3)
plt.plot(x,y,marker='o')
plt.show()
```



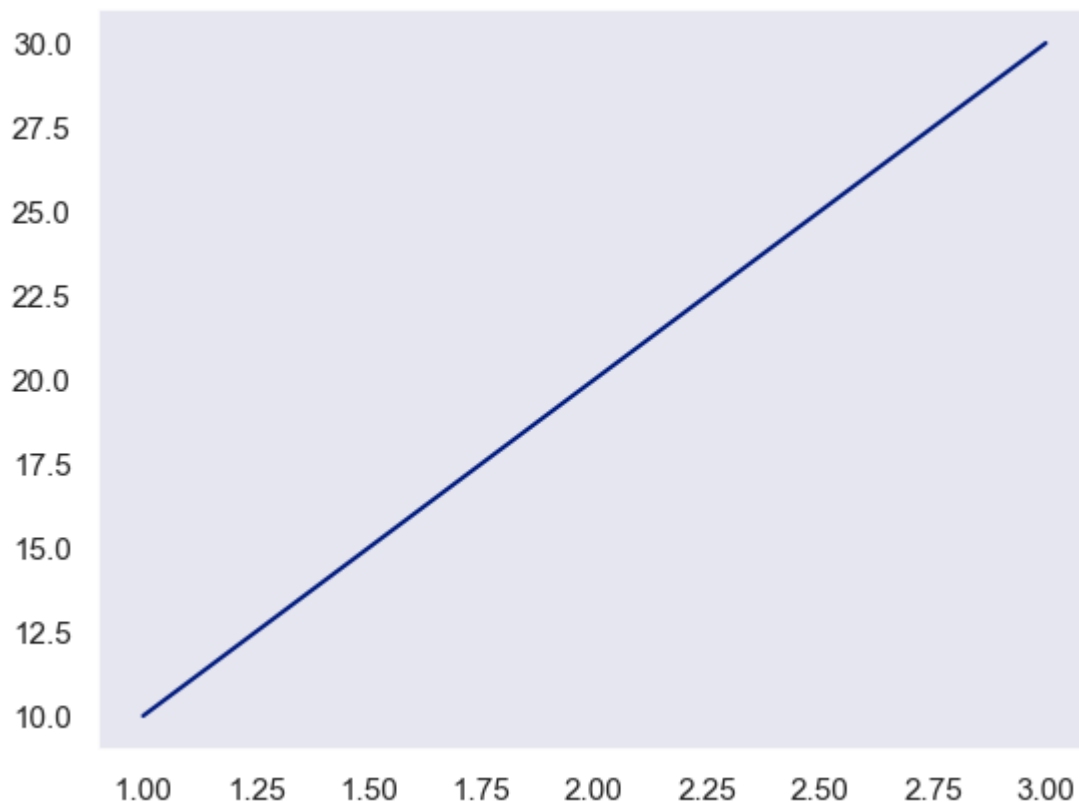
```
In [14]: !pip install seaborn
```

Requirement already satisfied: seaborn in c:\users\hp\anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\hp\anaconda3\lib\site-packages (from seaborn) (2.3.5)
Requirement already satisfied: pandas>=1.2 in c:\users\hp\anaconda3\lib\site-packages (from seaborn) (2.3.3)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\hp\anaconda3\lib\site-packages (from seaborn) (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\hp\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\hp\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\hp\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\hp\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)

```
In [22]: import seaborn as sns
import matplotlib.pyplot as plt
x = [1,2,3,4,5]
y = [10,20,30,40,50]
#Themes in seaborn
#theme control the appereance of Charts
#default theme
sns.set_theme()
#dark grid theme
sns.set_style('dark')
sns.lineplot(x=x,y=y)
plt.show()
```



```
In [29]: #color palette is a collection of color
sns.set_palette("dark")
sns.lineplot(
    x = [1,2,3],
    y = [10,20,30]
)
plt.show()
#available style
#deep
#bright
#muted
#pastel
#dark
#colorblind
```



```
In [31]: #built-in dataset support
#one
import seaborn as sns
print(sns.get_dataset_names())
```

```
['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes', 'diamonds', 'dots', 'dowjones', 'exercise', 'flights', 'fmri', 'geyser', 'glue', 'healthexp', 'iris', 'mpg', 'penguins', 'planets', 'seaice', 'taxis', 'tips', 'titanic']
```

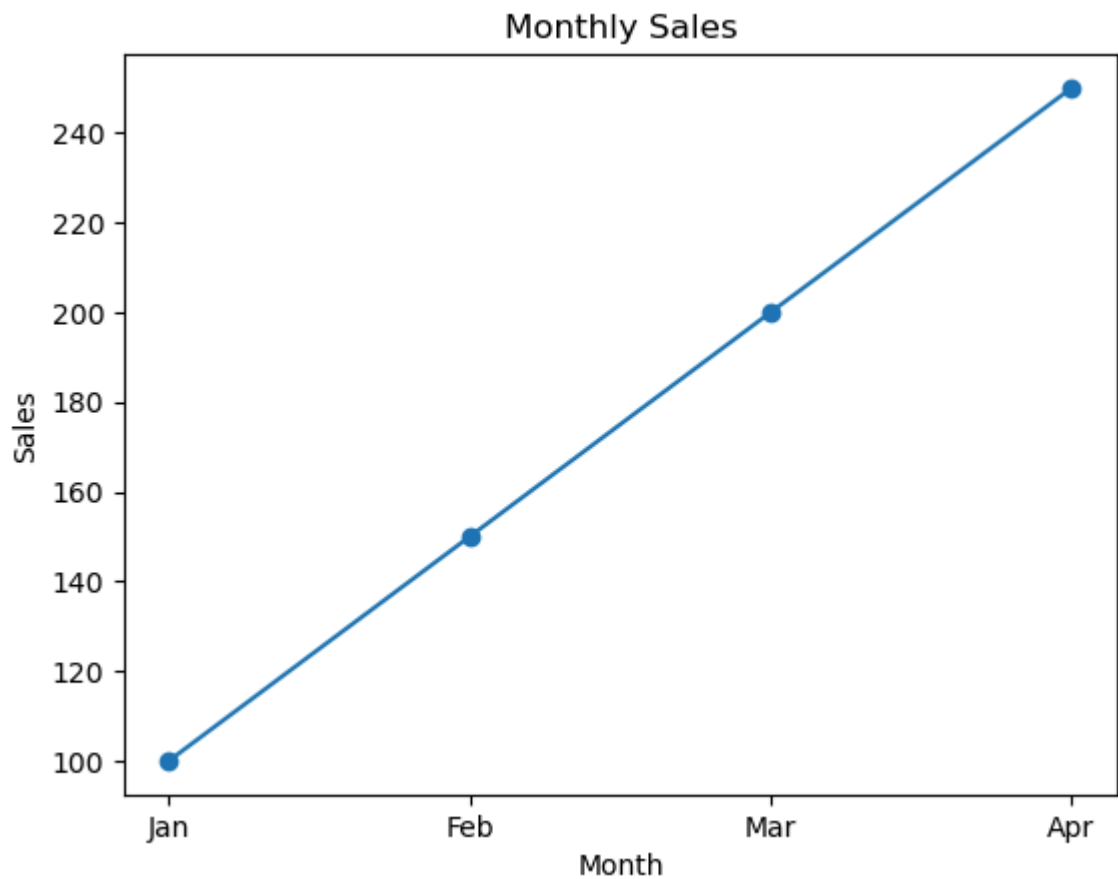
```
In [4]: #load titanic as sns
import seaborn as sns
df = sns.load_dataset("titanic")
print(df.head())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class
0	0	3	male	22.0	1	0	7.2500	S	Third
1	1	1	female	38.0	1	0	71.2833	C	First
2	1	3	female	26.0	0	0	7.9250	S	Third
3	1	1	female	35.0	1	0	53.1000	S	First
4	0	3	male	35.0	0	0	8.0500	S	Third

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

```
In [33]: import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr']
sales = [100, 150, 200, 250]
plt.plot(months, sales, marker='o')
plt.title("Monthly Sales")
```

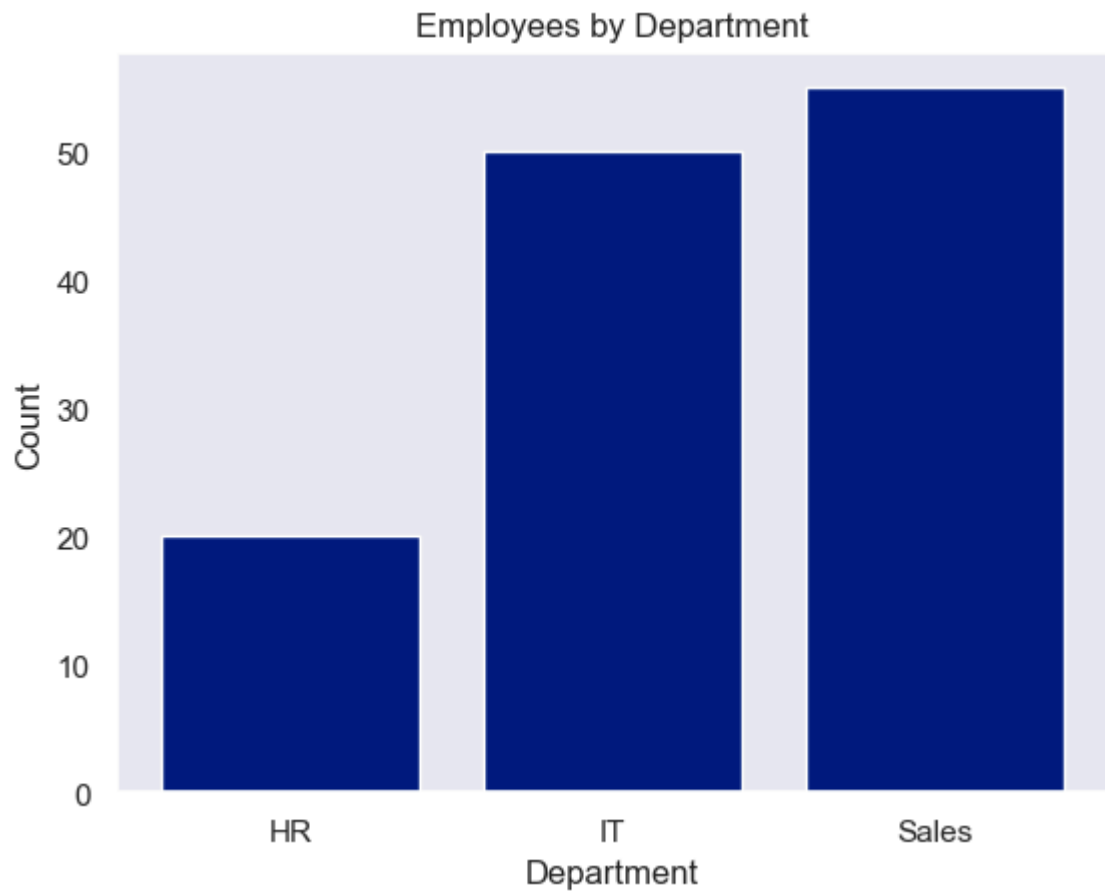
```
plt.xlabel("Month")
plt.ylabel("Sales")
#figure Size
plt.figure(figsize=(9,3))
plt.savefig("sales_chart.png")
plt.savefig(
    "chart.png",
    dpi=300
)
plt.show()
```



<Figure size 900x300 with 0 Axes>

BAR CHART

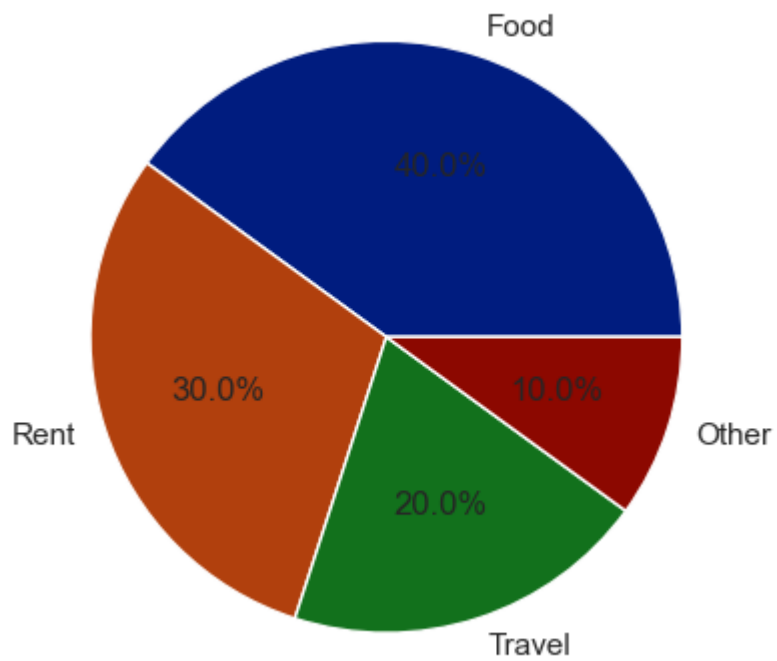
```
In [34]: import matplotlib.pyplot as plt
dept = ['HR', 'IT', 'Sales']
employees = [20, 50, 55]
plt.bar(dept, employees)
plt.title("Employees by Department")
plt.xlabel("Department")
plt.ylabel("Count")
plt.show()
```



Pie Chart

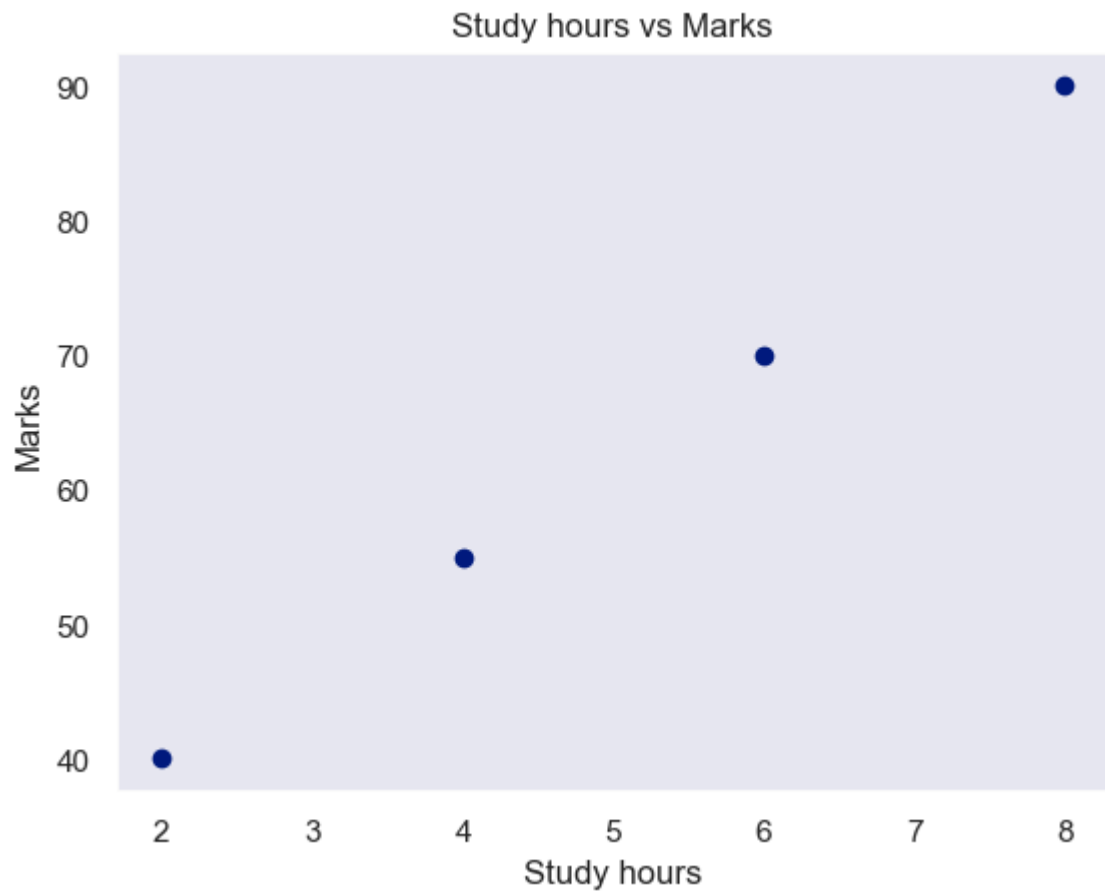
```
In [35]: import matplotlib.pyplot as plt
labels = ['Food', 'Rent', 'Travel', 'Other']
values = [40, 30, 20, 10]
plt.pie(
    values,
    labels = labels,
    autopct = '%1.1f%%'
)
plt.title("Monthly Expenses")
plt.show()
```

Monthly Expenses



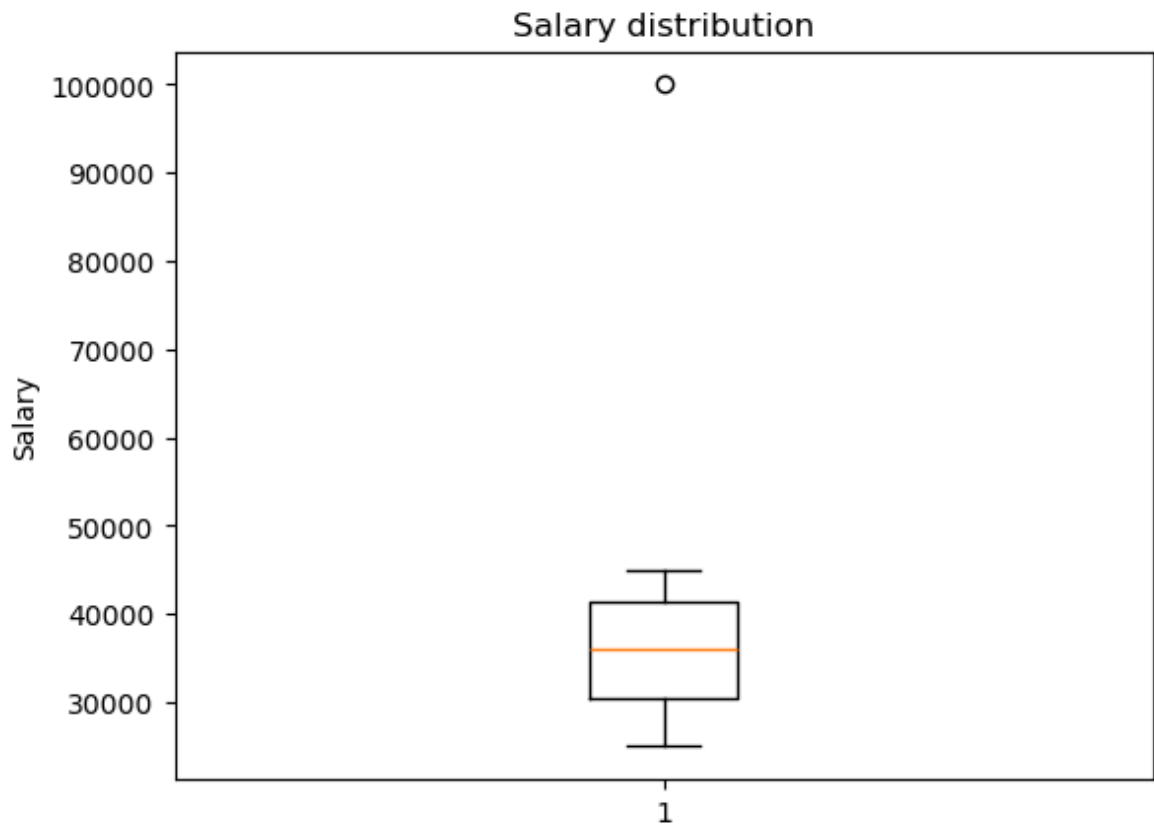
Scatter Plot

```
In [36]: import matplotlib.pyplot as plt
hours = [2,4,6,8]
marks = [40,55,70,90]
plt.scatter(hours, marks)
plt.title("Study hours vs Marks")
plt.xlabel("Study hours")
plt.ylabel("Marks")
plt.show()
```



Box Plot

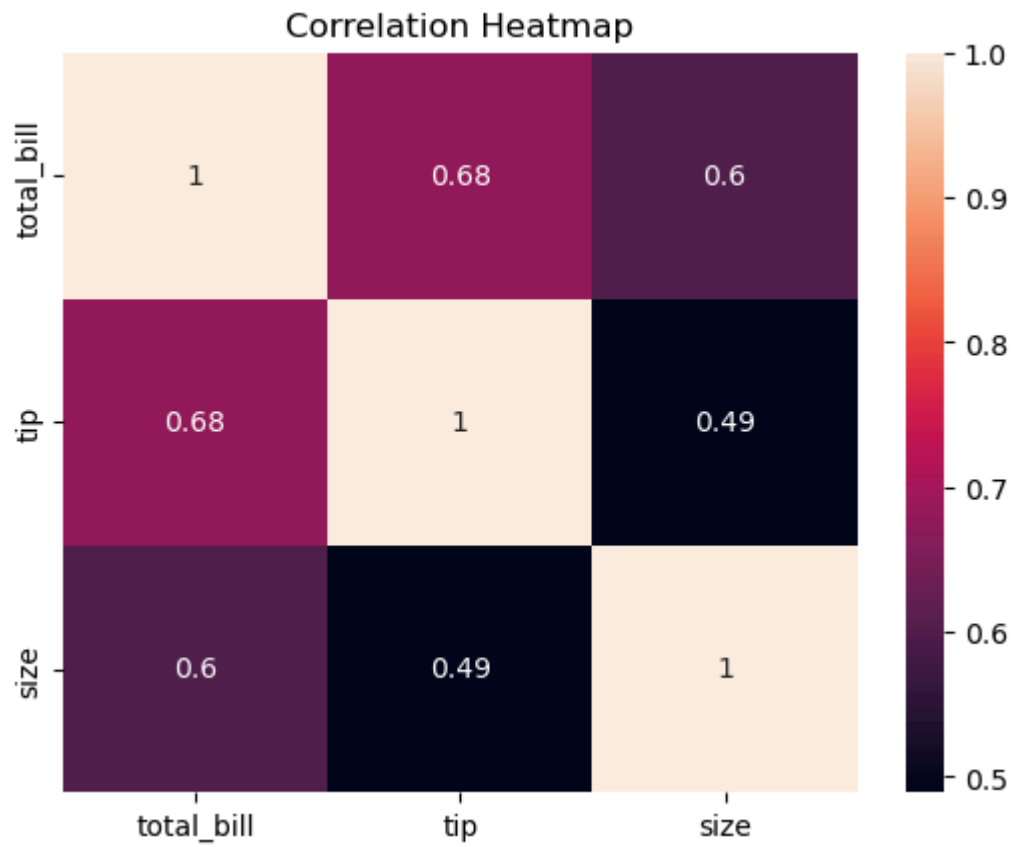
```
In [1]: import matplotlib.pyplot as plt
salary = [25000, 28000, 30000, 32000, 35000, 37000, 40000, 42000, 45000, 100000]
plt.boxplot(salary)
plt.title('Salary distribution')
plt.ylabel("Salary")
plt.show()
```



Heatmap

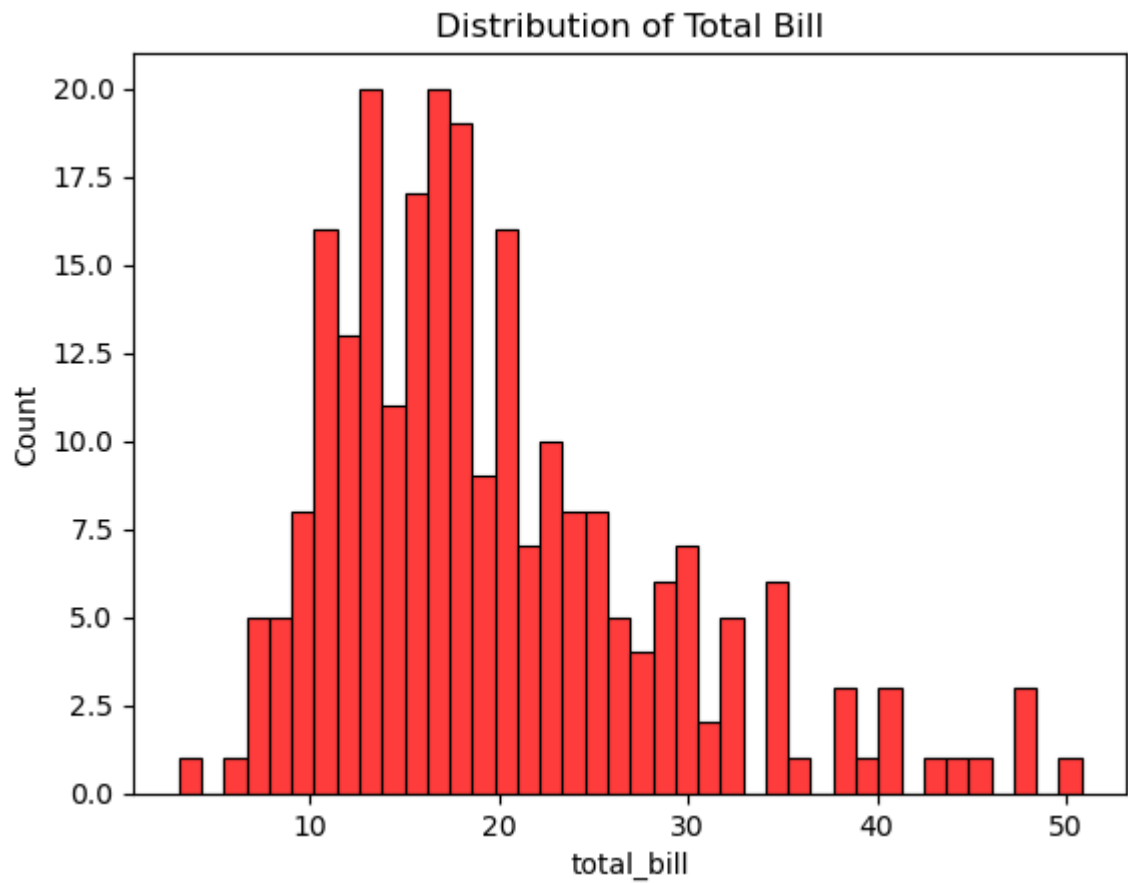
```
In [12]: import seaborn as sns
tips = sns.load_dataset("tips")
print(tips.head())
corr = tips.corr(numeric_only=True)
sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap")
plt.show()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4



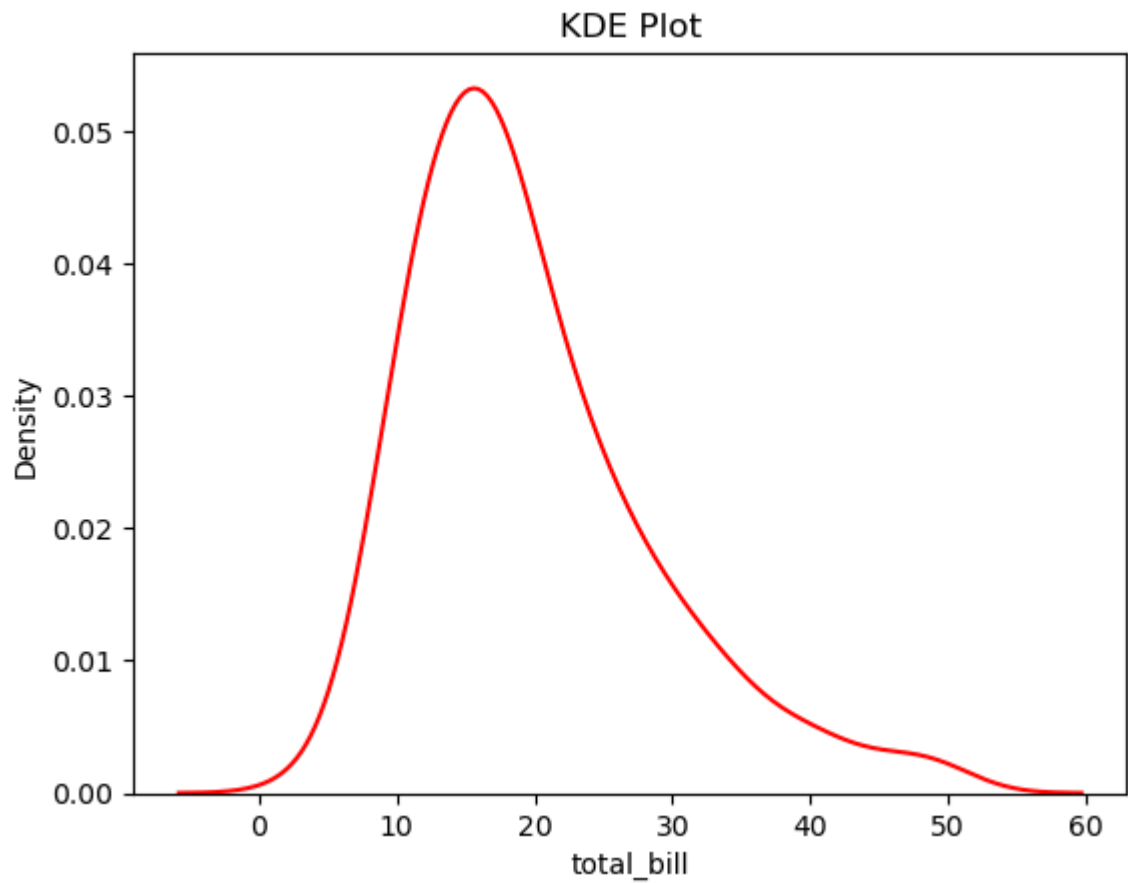
Histogram

```
In [17]: sns.histplot(  
    data=tips,  
    x="total_bill",  
    bins=40,  
    color="red"  
)  
plt.title("Distribution of Total Bill")  
plt.show()
```



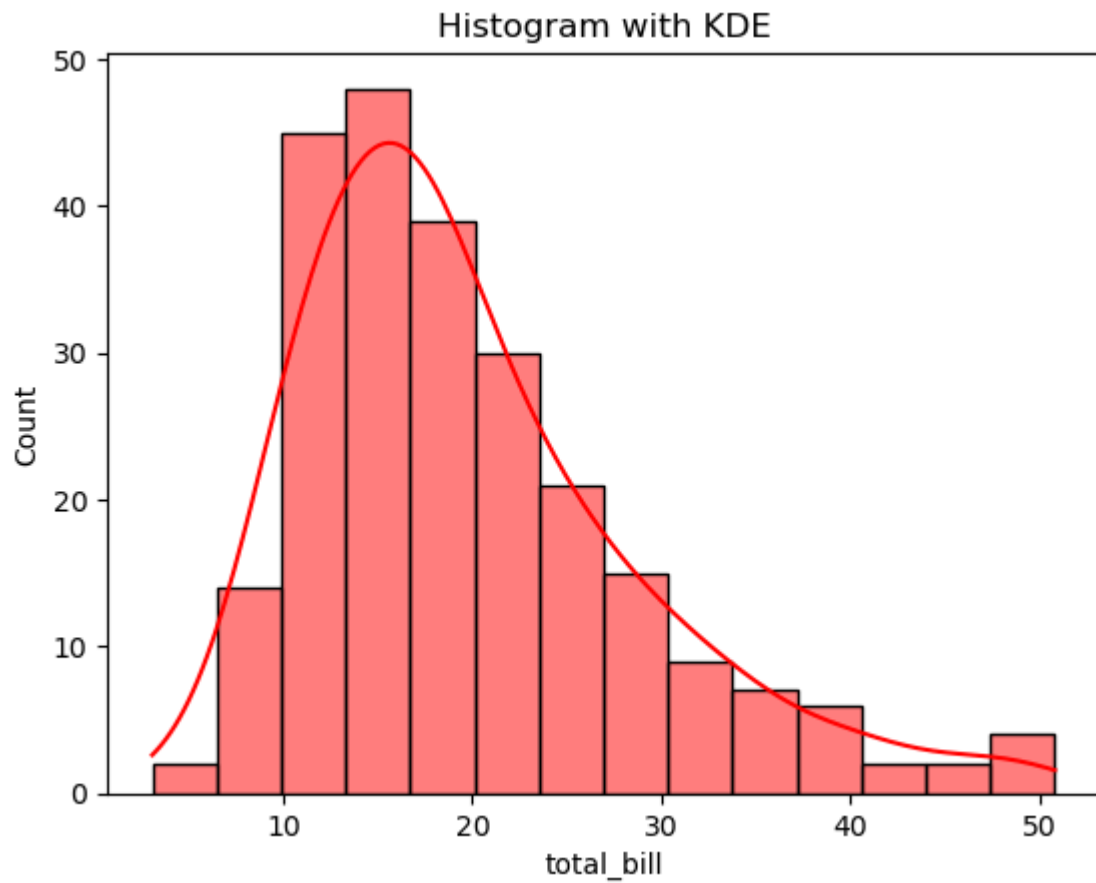
KDE Plot

```
In [19]: sns.kdeplot(  
    data=tips,  
    x="total_bill",  
    color="red"  
)  
plt.title("KDE Plot")  
plt.show()
```



Histogram + KDE

```
In [21]: sns.histplot(  
    data=tips,  
    x="total_bill",  
    color="red",  
    kde=True  
)  
plt.title("Histogram with KDE")  
plt.show()
```



```
In [28]: #2 charts

import matplotlib.pyplot as plt

plt.figure(figsize=(8,4))

# First Plot

plt.subplot(1,2,2)

plt.plot([1,2,3], [10,20,30])

plt.title("Line Chart")

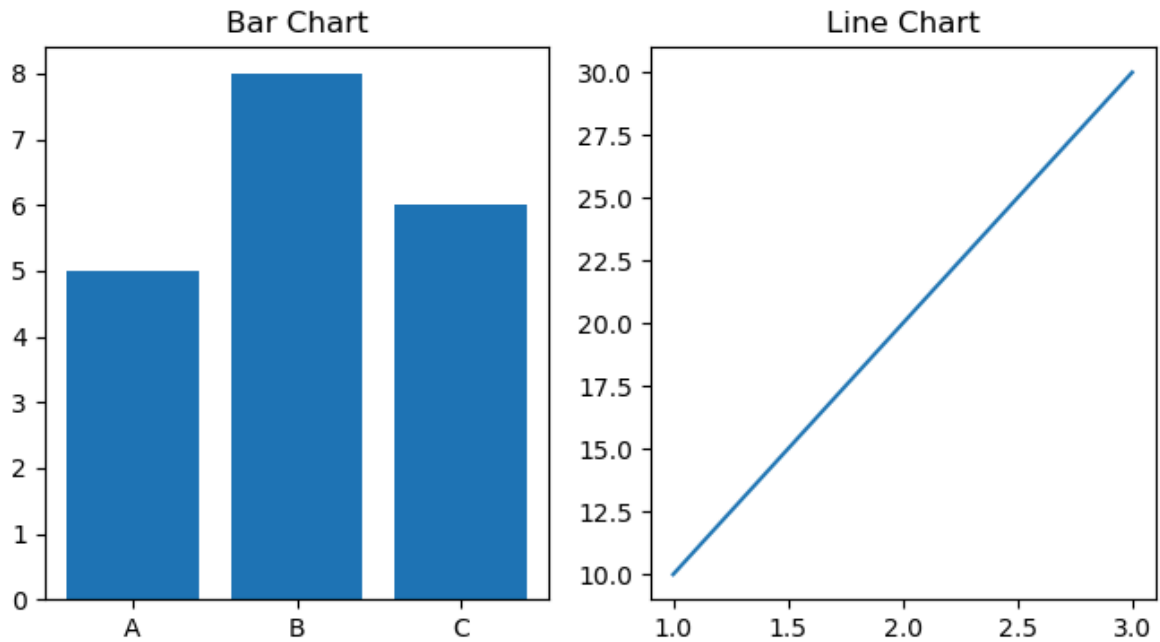
# Second Plot

plt.subplot(1,2,1)

plt.bar(['A','B','C'], [5,8,6])

plt.title("Bar Chart")

plt.show()
```



```
In [31]: #Example: Four Charts

import matplotlib.pyplot as plt

plt.figure(figsize=(10,8))

plt.subplot(2,2,1)

plt.plot([1,2,3], [10,20,30])

plt.title("Line")

plt.subplot(2,2,2)

plt.bar(['A','B','C'], [5,8,6])

plt.title("Bar")

plt.subplot(2,2,3)

plt.scatter([1,2,3], [4,8,6])

plt.title("Scatter")

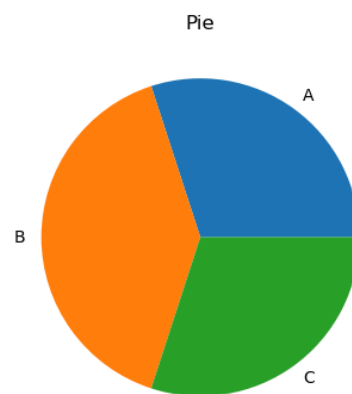
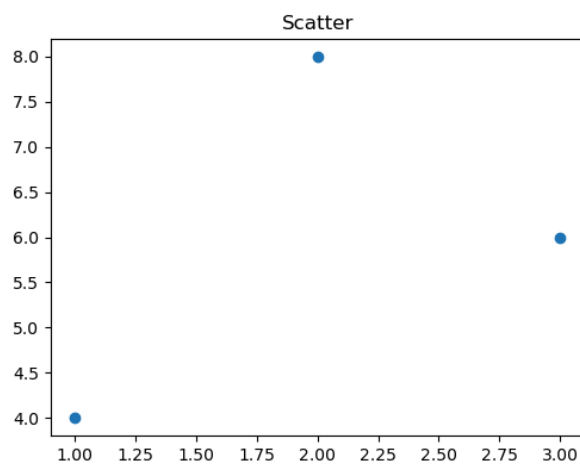
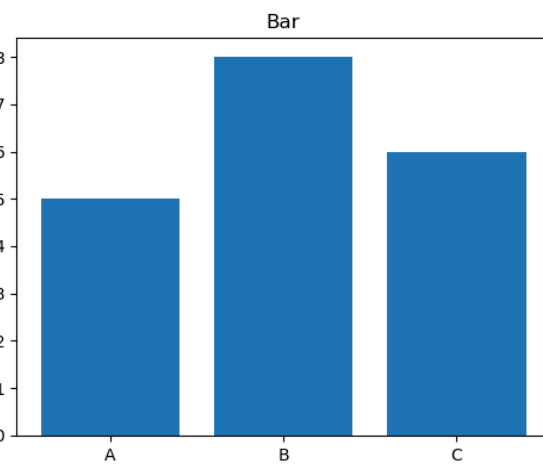
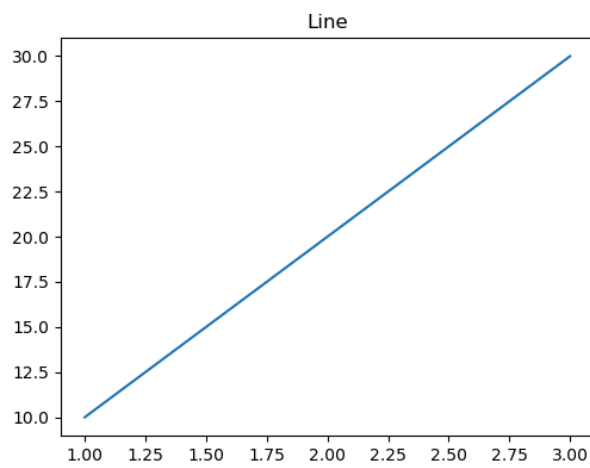
plt.subplot(2,2,4)

plt.pie([30,40,30], labels=['A','B','C'])

plt.title("Pie")

plt.tight_layout()

plt.show()
```



In []: